Weekly Progress Log – Week 3

Project Title: Data Science with Python
Created By: Preshna Acharya
Week: 3

Objectives for Week

- Understand the difference between Python lists and NumPy arrays and why NumPy is useful for numerical computing.
- Learn how to create and manipulate one-dimensional and two-dimensional NumPy arrays.
- Practice NumPy indexing, slicing, Boolean indexing, broadcasting, and vectorized operations.
- Learn how to load and explore CSV datasets using Pandas.
- Practice cleaning and transforming tabular data using Pandas.
- Handle missing values and duplicate records in a dataset.
- Practice filtering rows using single and multiple conditions.
- Use groupby() to summarize and analyze data.
- Learn how to merge multiple DataFrames using a common column.
- Transform a raw dataset into an analysis-ready dataset and save it as a cleaned CSV file.

Work Completed

- Completed a NumPy exercise notebook covering 1D and 2D array creation, zeros, ones, arange(), linspace(), array properties, indexing, slicing, Boolean indexing, broadcasting, vectorized arithmetic, mathematical functions, aggregate operations, and row/column calculations using axis.
- Loaded the public Titanic CSV dataset into Pandas and explored it using head(), shape, columns, dtypes, info(), and describe().
- Examined the raw Titanic dataset containing 891 rows and 12 columns.
- Checked missing values and identified 177 missing Age values, 687 missing Cabin values, and 2 missing Embarked values.
- Filled missing Age values with the median, filled missing Embarked values with the mode, and removed the Cabin column because of its large number of missing values.
- Checked for duplicate rows and used drop_duplicates() to remove duplicates if present.
- Practiced filtering passengers by age, sex, survival status, and combined conditions.
- Practiced groupby() by calculating average age by sex, survival rate by passenger class, and passenger counts by embarkation port.
- Created a second DataFrame containing readable passenger class descriptions and merged it with the Titanic dataset using Pclass.
- Removed PassengerId and Ticket for the simple analysis and created a new FamilySize feature using SibSp, Parch, and the passenger themself.

- Standardized column names by converting them to lowercase and performed final validation of shape, missing values, duplicates, and data types.
- Produced an analysis-ready dataset containing 891 rows and 11 columns with no missing values and saved it as titanic_analysis_ready.csv.
- Organized the Week 3 work into separate NumPy and Pandas Jupyter Notebooks with comments and headings explaining the steps.

Challenges Faced

- Understanding the difference between Python lists and NumPy arrays.
- Understanding indexing and slicing in both 1D and 2D NumPy arrays.
- Understanding how broadcasting works between arrays of different shapes.
- Understanding vectorized operations and how they reduce the need for manual Python loops.
- Deciding how different types of missing values should be handled.
- Understanding why the median is appropriate for filling missing numerical values such as Age.
- Understanding how groupby() and aggregation work together.
- Understanding how two DataFrames can be merged using a common column.
- Understanding the difference between the raw dataset and the final analysis-ready dataset.
- Understanding how a derived feature such as FamilySize can add useful information for analysis.

Solutions Implemented

- Practiced NumPy operations with small arrays before moving to more complex examples.
- Used clear examples to understand indexing, slicing, Boolean indexing, broadcasting, and vectorized operations.
- Explored the Titanic dataset before cleaning to identify missing values, column types, and overall structure.
- Used the median to replace missing Age values and the mode to replace missing Embarked values.
- Removed the Cabin column because most of its values were missing.
- Used drop_duplicates() to ensure duplicate records were removed.
- Practiced multiple filtering conditions to understand Boolean filtering in Pandas.
- Used groupby() with mean() and size() to summarize grouped data.
- Created a passenger-class lookup DataFrame and merged it with the main dataset using Pclass.
- Created the FamilySize feature from existing columns and performed final checks before exporting the cleaned dataset.
- Saved the final cleaned dataset separately so the original raw CSV remained unchanged.

Individual Tasks

Responsibility	Description of Contribution	Remark
NumPy Array	Created 1D and 2D arrays and practiced zeros(), ones(),	Improved understanding

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Creation	arange(), and linspace().	of NumPy array creation.
NumPy Indexing and Slicing	Practiced indexing, slicing, 2D indexing, and Boolean indexing.	Improved ability to select and manipulate array values.
NumPy Broadcasting	Applied scalar broadcasting and broadcasting between 1D and 2D arrays.	Developed understanding of operations across different array shapes.
Vectorized Operations	Performed arithmetic and mathematical operations directly on NumPy arrays.	Learned efficient calculations without manual loops.
Aggregate Operations	Calculated sum, mean, minimum, maximum, standard deviation, and row/column statistics.	Practiced numerical data summarization.
Dataset Loading	Loaded the Titanic CSV dataset into Pandas.	Successfully imported a public dataset for analysis.
Data Exploration	Examined dataset shape, columns, data types, info, and statistical summaries.	Improved understanding of dataset structure.
Missing Value Handling	Filled missing Age and Embarked values and removed Cabin.	Dataset cleaned of missing values.
Duplicate Handling	Checked for and removed duplicate records.	Improved dataset consistency.
Data Filtering	Filtered passengers by age, sex, survival status, and combined conditions.	Practiced Boolean row filtering.
GroupBy Analysis	Calculated average age, survival rate, and passenger counts using groupby().	Practiced grouped analysis and aggregation.
Data Merging	Created a passenger-class DataFrame and merged it with the Titanic dataset.	Learned how to combine datasets using a common key.
Feature Engineering	Created the FamilySize column using SibSp and Parch.	Created a useful new variable from existing data.
Data Cleaning	Removed unnecessary columns and standardized column names.	Prepared the dataset for further analysis.
Final Validation	Checked final shape, missing values, duplicates, and data types.	Confirmed that the dataset was analysis-ready.
Dataset Export	Saved the cleaned dataset as titanic_analysis_ready.csv.	Final cleaned dataset prepared for future analysis.
Documentation	Organized NumPy and Pandas work in Jupyter Notebooks with comments and headings.	Week 3 work documented and prepared

Responsibility	Description of Contribution	Remark
		for submission.